Jacob MacInnis

Vice President of Engineering | Cloud & AI

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SUMMARY

Hand-on engineering leader, and top IC, architecting scalable cloud-native distributed systems and building high-performing teams. Proven track record of fast, reliable shipping, optimizing engineering processes, driving major feature development and system migrations, improving system reliability and ensuring long-term velocity and quality across the full software development lifecycle.

Combines technical expertise with strong leadership, clear communication, and strategic vision to drive innovation in remote-first environments. Known for a thoughtful, creative approach to complex engineering challenges. Promotes collaboration, continuous improvement and fosters pragmatic team cultures delivering intuitive solutions.

Applied AI/ML experience across NLP, LLMs, classification, image recognition, RAG, and agentic workflows. Proficient in TensorFlow, PyTorch, LangChain, and LangGraph, with certifications from Stanford, DeepLearning.AI, Duke, and Google, combining theory with real-world delivery.

KEY SKILLS

Frontend: React, Redux, TypeScript, JavaScript
Backend: Node.js, Python, FastAPI
Cloud: AWS, GCP, Azure, Serverless Framework
ML/AI: TensorFlow, PyTorch, LangChain, LangGraph, scikit-learn
Data: Firestore, MongoDB, PostgreSQL
DevOps: Docker, Circle CI, Terraform
Leadership: Team Management, Coaching, Mentorship, Product Vision, Process Optimization

EXPERIENCE

10/2026 - Present **Vice President of Engineering, OneView Commerce** Remote/Boston

- Head of the global Engineering and QA organizations, driving engineering practice, product delivery, and platform quality across OneView's headless unified commerce engine.
- Manage engineering teams across global time zones, developing technical leaders, balancing seniority depth, and partnering closely with Product, Architecture, and enterprise client executives.
- Spearheading AI adoption across development and QA lifecycles, integrating LLM context, agent skills, and token optimization into the monorepo, increasing delivery velocity and engineer scope while enforcing strict technical accountability.
- Oversee cloud-native, event-driven, edge-to-cloud architectures engineered for high-availability physical retail and complex network environments.
- Merged QA into a single engineering organization, embedding automated release gates and shifting quality ownership directly to software engineering teams.

06/2022 - 10/2026 **Director of Engineering, OneView Commerce** Remote/Boston

- Implemented global-scale development of microservice cloud infrastructure, supporting over 10,000 retail stores and processing millions transactions daily.
- Optimized deployment processes, significantly reducing downtime from several days to just 1-2 hours with a remarkable 99% success rate.
- Engineered a robust serverless architecture using AWS, which included over 300 Cloud Functions (Lambda, SQS, SNS, API Gateway), GCP Functions, and Firestore DB.

- Directed the successful migration of legacy databases to Firestore, effectively reducing production database issues by over 100%.
- Championed culture of systems thinking, continuous improvement in code editing, collaboration and consistent architectural practices, enabling engineers to anticipate downstream impacts across cloud infrastructure and APIs and improve product business value.
- Led 3+ cross-functional teams (12+ engineers), driving execution, performance management, engineering quality, and team performance across a globally scaled platform.
- Architected and delivered a semi-autonomous agentic chatbot for corporate retail users, enabling self-service transaction data retrieval, search, and lightweight analysis and anomaly detection.
- Drove adoption of AI as a development force multiplier by integrating LLM context, agent skills and token reduction optimization into monorepo and upskilling engineers on AI-assisted workflows.

03/2021 - 06/2022 **Engineering Manager, OneView Commerce** Remote/Boston

- Architected and spearheaded the development of a globally deployed order, pick, and pack platform utilized by major retailers, featuring a cloud-native backend utilizing AWS, GCP, and Firestore, along with a React/Redux frontend.
- Formulated a multicloud backend monorepo, creating a cloud-agnostic platform that facilitated AWS cloud and Azure deployments while enabling future support for GCP.
- Led two cross-functional teams, managing over 8 engineers and overseeing full-stack development encompassing React, Node, TypeScript/JavaScript, and serverless system architecture.
- Mentored engineers in system design, problem-solving skills, code editing, debugging, and delivery best practices. Honing their product understanding, and confidence under pressure as well as verbal and written communication skills particularly their ability to articulate technical concepts to stake holders and non technical teams.
- Drove architectural enablement initiatives across 4 software development teams, standardizing platform practices, collaboration and accelerating knowledge transfer among product teams.

07/2020 - 03/2021 **Team Lead, Software Engineering, OneView Commerce** Remote/Boston

- Led a scrum team building headless SaaS apps with a focus on innovation and fast iteration
- Developed OneView's platform user management infrastructure, analogous to AWS IAM's users, roles, policies, and permissions, utilizing AWS Cognito user pool.
- Revamped development team meetings by introducing structured agendas and focused discussions, data-driven approach to cloud optimization, which improved cross-team alignment, collaboration and momentum.
- Developed an enterprise search solution for the Point of Sale mobile and desktop application as well as the administration dashboard web app using Typesense search engine.
- Built scalable backend services with AWS Lambda, S3, API Gateway, and Firestore for fast delivery.

08/2019 - 07/2020 **Full Stack Software Engineer, OneView Commerce** Remote/Boston

- Engineered a cloud-agnostic integration layer that expedited product development cycles and mitigated technical debt through a unified codebase for multi-cloud deployment.
- Conceptualized a wrapper interface that abstracted common cloud service concepts, dynamically mapping them to AWS, Azure, and prospective cloud providers.
- Assessed + fortified penetration test security measures, addressing vulnerabilities such as XSS and function-level access controls while improving API security key management.

02/2019 - 08/2019 **Full Stack Engineer, Insight Global** Remote/Boston

- Developed a curbside pickup solution for Kroger, facilitating online purchases for pickup at designated locations such as Kroger and Walgreens stores.
- Built SaaS and headless POS features, incl. barcode handling, and mobile USB scanning.
- Translated business requirements into agile epics with sprint-sized, vertically sliced increments.

11/2018 - 04/2019 **Architect & Full Stack Developer, Function Earth (self start-up)** Remote/Boston

- Constructed cross-platform React Native mobile applications for both iOS and Android devices.
- Integrated JavaScript Node.js backend services with RESTful API endpoints and MongoDB.

05/2018 - 11/2018 **Developer Apprentice, Thinkful** Remote/Boston

- Deployed mobile-first responsive web applications using JavaScript Node, Express RESTful API, React.js, Redux, and databases such as Postgres and Mongo.

05/2015 - 07/2018 **Founder & President, E-Prime Sourcing** Salem, MA

- Founded and scaled E-Prime Sourcing into a commercial warehouse, generating $1.3M+ in sales.
- Secured a $3M distribution deal and fulfilled 100K+ orders across Amazon, Walmart, and eBay.
- Built a Shopify site with 350+ products and listed 800+ SKUs across marketplaces.

EDUCATION

University of Colorado Boulder • Masters of Science, Artificial Intelligence
(In Progress)

Stanford • Certificate, Machine Learning Specialization

- supervised/unsupervised/reinforcement learning, model evaluation, and tuning best practices.
- neural networks, regression, clustering, and ML model optimization taught by Andrew Ng.

Duke • Certificate, Managing Machine Learning Projects

- applied CRISP-DM framework, managed ML project lifecycle, and aligned teams for delivery.
- assess ML feasibility, manage data pipelines, and design production monitoring strategies.

DeepLearning.AI • Certificate, Machine Learning in Production

- covered deployment strategies, data drift detection, and model lifecycle management.
- built skills in packaging ML models, setting up monitoring, and maintaining production pipelines.

Thinkful • Certificate, Software Engineering

- Completed 1,600 hours of intensive training in full-stack software engineering, covering core concepts in computer science and web development.

Eastern Nazarene College • Bachelors, Child Psychology

SELECTED AI / MACHINE LEARNING PROJECTS

- MCP App – Explored MCP-style orchestration: with LangChain and LangGraph
- R2AI – LangGraph agent that routes user intent across tools (search, math, code)
- Agent Cortex – LangChain AI Agent with short-term memory and FAISS-based long-term vector memory. Tools include RAG, search, LLaMA/Mistral or OpenAI.
- RAG API – *FastAPI backend with RAG (Retrieval Augmented Generation) + FLAN-T5 API*
- Summit Mind – Document Summarization, fine-tune neural networks. Deploy with Docker, Artifact Registry, CloudRun
- Folia-Vision – Plant Disease image recognition, Transfer Learning with TensorFlow/Keras
- Career Compass – Employment Predict classification TensorFlow/Keras neural networks

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KEY SKILLS

Tools: Pandas, NumPy, Matplotlib, Seaborn
Data Skills: Data Visualization, Data Analysis, Cleaning & Wrangling, Preprocessing
AI/ML Concepts: Model Development, AI/ML Tech, Innovation Strategies
Leadership/Architecture: Technical Guidance, Accountability Structures, Architectural Patterns